

Full Completion Additional Experiment Tables

Purpose. This table package replaces the prior Gisette placeholder coverage rows with numeric rerun results where the completion artifacts exist. It keeps mixed results visible instead of turning every row into a positive narrative.

Key caveats. Gisette lambda/seed rows are fixed-fold seed reruns, the Gisette sequential row is a fixed-fold schedule comparison with a mixed outcome, and the Gisette correlated-feature row remains a representative top-100 subset stress test.

Coverage Matrix

evidence_block	bcwd	vowel	spambase	gisette	claim_boundary
Threshold tau sweep	numeric	numeric	numeric	numeric	4-dataset sensitivity evidence
Sparsity lambda/seed retraining	numeric	numeric	numeric	numeric	Gisette row is fixed fold 1, seeds 0-4; avoid claiming 5-fold lambda sweep
Full vs Primary checkpoint attribution	numeric	numeric	numeric	numeric	4-dataset Full-Primary delta and zero hard-route overlap
Sequential vs joint schedule	numeric	numeric	numeric	numeric	Gisette is numeric but mixed; no universal sequential-superiority claim
Architecture training-tier ablation	numeric	numeric	numeric	numeric	Use for attribution; preserve cases where dense/no-disjoint is higher
Correlated-copy stress	numeric	numeric	numeric	numeric top-100 subset	Group-level explanation under high collinearity
Product-vs-HTSK firing comparison	numeric	numeric	numeric	numeric	HTSK as numerical scale-control, not sole performance cause
Strong tabular baselines	numeric	numeric	numeric	numeric	Do not claim universal dominance over all tabular baselines
SHAP/LIME local deletion	numeric	numeric	numeric	numeric	GRS explanations are complementary, not universally more faithful locally
Intrinsic rule-path metrics	numeric	numeric	numeric	numeric	Model-native rule-path evidence
Paired statistics	numeric	numeric	numeric	numeric	Current revision-rerun fold evidence only

Threshold Sensitivity

dataset	tau_0.3_f1	tau_0.4_f1	tau_0.5_f1	tau_0.6_f1	tau_0.7_f1	selected_a_t_0.5	mask_jaccard_at_0.5	source
Breast Cancer	0.9415	0.9542	0.9524	0.9412	0.9267	75.8	0.9045	MAT-5 legacy/revision robustness summary
Vowel	0.9394	0.9524	0.9606	0.9334	0.8868	21.0	0.7909	MAT-5 legacy/revision robustness summary
Spambase	0.8842	0.9009	0.9120	0.8658	0.7941	48.0	0.8056	MAT-5 legacy/revision robustness summary
Gisette	0.9757	0.9757	0.9758	0.6667	0.6667	2680.80	0.6011	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_completion/artifacts/

Sparsity Lambda Seed

dataset	lambda	full_acc	full_f1	selected_f eatures	mask_ja ccard	pc_overl ap	source
Breast Cancer	0	0.9714 +/- 0.0101	0.9567 +/- 0.0156	69.2 +/- 3.2	0.8102	0.0000	MAT-5 targeted rerun
Breast Cancer	0.01	0.9771 +/- 0.0083	0.9658 +/- 0.0126	40.6 +/- 2.7	0.7334	0.0000	MAT-5 targeted rerun
Breast Cancer	0.1	0.9600 +/- 0.0035	0.9402 +/- 0.0049	13.2 +/- 1.3	0.6486	0.0000	MAT-5 targeted rerun
Vowel	0	0.9616 +/- 0.0118	0.9614 +/- 0.0119	26.8 +/- 1.3	0.9152	0.0000	MAT-5 targeted rerun
Vowel	0.01	0.9545 +/- 0.0090	0.9544 +/- 0.0091	21.4 +/- 3.1	0.7880	0.0000	MAT-5 targeted rerun
Vowel	0.1	0.9394 +/- 0.0146	0.9387 +/- 0.0146	10.0 +/- 0.0	1.0000	0.0000	MAT-5 targeted rerun
Spambase	0	0.9309 +/- 0.0023	0.9117 +/- 0.0033	47.4 +/- 1.4	0.8778	0.0000	MAT-5 targeted rerun
Spambase	0.01	0.9262 +/- 0.0045	0.9055 +/- 0.0054	26.8 +/- 2.3	0.7707	0.0000	MAT-5 targeted rerun
Spambase	0.1	0.9058 +/- 0.0064	0.8776 +/- 0.0077	9.2 +/- 0.4	0.9000	0.0000	MAT-5 targeted rerun
Gisette	0.0	0.9752 +/- 0.0008	0.9751 +/- 0.0008	4543.8 +/- 26.9	0.8761	0.0000	/home/harp3133t/Research/ 03_Research/GH-ANFIS_E404/ rebuttal_full_completion/artifacts/ gisette_full_completion/ mat5_threshold_sparsity_seed_rerun/ lambda_seed_summary.csv
Gisette	0.01	0.9760 +/- 0.0010	0.9759 +/- 0.0010	2399.6 +/- 12.3	0.8557	0.0000	/home/harp3133t/Research/ 03_Research/GH-ANFIS_E404/ rebuttal_full_completion/artifacts/ gisette_full_completion/ mat5_threshold_sparsity_seed_rerun/ lambda_seed_summary.csv
Gisette	0.1	0.9692 +/- 0.0020	0.9691 +/- 0.0021	358.0 +/- 5.9	0.8170	0.0000	/home/harp3133t/Research/ 03_Research/GH-ANFIS_E404/ rebuttal_full_completion/artifacts/ gisette_full_completion/ mat5_threshold_sparsity_seed_rerun/ lambda_seed_summary.csv

Full Vs Primary Checkpoint

dataset	full_f1	primary_f1	delta	effective_overlap	source
Breast Cancer	0.9630	0.9585	+0.0044	0.0	MAT-6 checkpoint attribution
Vowel	0.9662	0.9651	+0.0011	0.0	MAT-6 checkpoint attribution
Spambase	0.9136	0.9059	+0.0076	0.0	MAT-6 checkpoint attribution
Gisette	0.9758	0.9755	+0.0003	0.0	MAT-6 checkpoint attribution

Sequential Vs Joint

dataset	sequential_f1	joint_no_freeze_f1	stage1_unfreeze_f1	effective_overlap	claim_boundary
Breast Cancer	0.9373	0.9171	0.9288	0.0	numeric schedule comparison
Vowel	0.9708	0.9347	0.9696	0.0	numeric schedule comparison
Spambase	0.9191	0.8045	0.9056	0.0	numeric schedule comparison
Gisette	0.9732	0.9757	0.9784	0.0000	numeric fixed-fold rerun; mixed result, so no universal sequential-superiority claim

Architecture Ablation

dataset	variant	variant_label	full_acc	full_f1	selected_features	mask_overlap_rate	primary_f1	full_minus_primary_f1	source
Breast Cancer	full_grs	Full GRS-ANFIS	0.9556 +/- 0.0139	0.9338 +/- 0.0226	72.20	0.0000	0.9246	0.0092	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv
Breast Cancer	primary_only_same_budget	Primary-only, same epoch budget	0.9528 +/- 0.0097	0.9308 +/- 0.0150	37.00	0.0000	0.9308	0.0000	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_co

Breast Cancer	joint_training_same_budget	Joint training, same epoch budget	0.9528 +/- 0.0097	0.9304 +/- 0.0149	52.60	0.0000	0.9263	0.0041	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv
Breast Cancer	no_disjoint	No disjoint complementary routing	0.9499 +/- 0.0064	0.9260 +/- 0.0115	73.80	0.4565	0.9257	0.0003	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv
Breast Cancer	no_hard_finetune_same_budget	No hard fine-tune, same epoch budget	0.9413 +/- 0.0083	0.9129 +/- 0.0143	78.60	0.0000	0.8957	0.0172	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/

									rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv/home/harp3133t/Research/03_Research/GH-ANFIS_E404/
Breast Cancer	dense_tsk_same_rules	Dense TSK-ANFIS, same total rules/epochs	0.9628 +/- 0.0114	0.9462 +/- 0.0167	80.00				rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv/home/harp3133t/Research/03_Research/GH-ANFIS_E404/
Breast Cancer	prod_firing	Product firing instead of HTSK	0.9514 +/- 0.0199	0.9286 +/- 0.0293	78.40	0.0000	0.9291	-0.0004	rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_bcwd/attribution_summary.csv/home/harp3133t/Research/03_Research/GH-ANFIS_E404/
Gisette	full_grs	Full GRS-ANFIS	0.9705 +/- 0.0032	0.9704 +/- 0.0032	2698.00	0.0000	0.9701	0.0003	/home/harp3133t/Research/03_Research/GH-

									ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv /home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv /home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv
Gisette	primary_ only_sam e_budget	Primary-only, same epoch budget	0.9710 +/- 0.0040	0.9710 +/- 0.0039	2287.00	0.0000	0.9710	0.0000	ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv /home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv /home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv
Gisette	joint_trai ning_sam e_budget	Joint training, same epoch budget	0.9687 +/- 0.0037	0.9686 +/- 0.0036	3333.50	0.0000	0.9687	-0.0000	ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv /home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv
Gisette	no_disjoi nt	No disjoint complementary routing	0.9722 +/- 0.0042	0.9722 +/- 0.0041	2394.50	0.3207	0.9727	-0.0005	/home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_gisette/ attribution_sum mary.csv

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Gisette	no_hard_ finetune_ same_bu dget	No hard fine- tune, same epoch budget	0.9675 +/- 0.0038	0.9674 +/- 0.0038	2346.50	0.0000	0.9669	0.0005	
Gisette	dense_tsk _same_ru les	Dense TSK- ANFIS, same total rules/epochs	0.9685 +/- 0.0045	0.9684 +/- 0.0045	5000.00				
Gisette	prod_firi ng	Product firing instead of HTSK	0.5000 +/- 0.0000	0.3333 +/- 0.3333	0.00		0.6667	-0.3333	/home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/

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Spambase	full_grs	Full GRS-ANFIS	0.9352 +/- 0.0058	0.9172 +/- 0.0080	56.00	0.0000	0.9092	0.0080	
Spambase	primary_only_same_epoch_budget	Primary-only, same epoch budget	0.9333 +/- 0.0081	0.9148 +/- 0.0102	41.20	0.0000	0.9148	0.0000	
Spambase	joint_training_same_epoch_budget	Joint training, same epoch budget	0.8570 +/- 0.0481	0.8336 +/- 0.0373	53.00	0.0000	0.8006	0.0331	/home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_completion/ artifacts/ gisette_full_completion/

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Spambase	no_disjoi nt	No disjoint complementary routing	0.9385 +/- 0.0069	0.9211 +/- 0.0096	52.40	0.3574	0.9034	0.0177	
Spambase	no_hard_ finetune_ same_bu dget	No hard fine- tune, same epoch budget	0.9294 +/- 0.0074	0.9099 +/- 0.0096	56.80	0.0000	0.8696	0.0403	
Spambase	dense_tsk _same_ru les	Dense TSK- ANFIS, same total rules/epochs	0.9418 +/- 0.0051	0.9257 +/- 0.0062	57.00				/home/ harp3133t/ Research/ 03_Research/ GH- ANFIS_E404/ rebuttal_full_co mpletion/ artifacts/ gisette_full_co mpletion/ mat6_architectu re_spambase/

Spambase	prod_firing	Product firing instead of HTSK	0.9220 +/-	0.9003 +/-	56.40	0.0000	0.9009	-0.0006	attribution_summary.csv
			0.0104	0.0143					/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttall_full_completion/artifacts/gisette_full_completion/mat6_architecture_spambase/attribution_summary.csv
Vowel	full_grs	Full GRS-ANFIS	0.9657 +/-	0.9656 +/-	27.40	0.0000	0.9584	0.0073	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttall_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_vowel/attribution_summary.csv
Vowel	primary_only_same_budget	Primary-only, same epoch budget	0.9556 +/-	0.9555 +/-	16.40	0.0000	0.9555	0.0000	/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttall_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_vowel/

Vowel	dense_tsk	Dense TSK-								re_vowel/
	_same_rules	ANFIS, same total rules/epochs	0.9727 +/- 0.0052	0.9726 +/- 0.0053	29.00					attribution_summary.csv
Vowel	prod_firing	Product firing instead of HTSK	0.9475 +/- 0.0185	0.9474 +/- 0.0185	28.20	0.0000	0.9370	0.0104		/home/harp3133t/Research/03_Research/GH-ANFIS_E404/rebuttal_full_completion/artifacts/legacy_preprocessing_bcwvowel/mat6_architecture_re_vowel/attribution_summary.csv

Correlated Feature Stress

dataset	acc_delta	selected_delta	raw_jaccard	collapsed_jaccard	role_assignment_summary	scope
Breast Cancer	0.0000	+3.2	0.8003	0.8207	same 64%, single 24%, neither 12%	full dataset
Vowel	+0.0091	+8.2	0.9102	0.8930	same 100%	full dataset
Spambase	+0.0017	0.0	0.7412	0.7970	same 28%, single 44%, neither 8%, split 20%	full dataset
Gisette	0.0000	+5.2	0.8745	0.8732	same 76%, single 4%, neither 20%	representative top-100 subset

Product Vs Htsk Firing

dataset	mode	acc	f1	base_near _zero	comp_nea r_zero	comp_max _weight	comp_eff ective_rules	source
Breast Cancer	HTSK	0.9557	0.9353	0.0000	0.0000	0.1032	10.94	MAT-9 product-vs- HTSK summary
Breast Cancer	product	0.9657	0.9501	0.1354	0.0000	0.1482	10.16	MAT-9 product-vs- HTSK summary
Gisette	HTSK	0.9733 +/- 0.0053	0.9733 +/- 0.0052	0.0000 +/- 0.0000	0.0000 +/- 0.0000	0.0914 +/- 0.0000	10.9999 +/- 0.0000	/home/harp3133t/ Research/03_Research/ GH-ANFIS_E404/ rebuttal_full_completion/ artifacts/ gisette_full_completion/ htsk_firing_vowel_gisett e/reviewer_table.csv
Gisette	prod	0.5000 +/- 0.0000	0.4444 +/- 0.3849	0.0000 +/- 0.0000	0.0000 +/- 0.0000	0.0909 +/- 0.0000	11.0000 +/- 0.0000	/home/harp3133t/ Research/03_Research/ GH-ANFIS_E404/ rebuttal_full_completion/ artifacts/ gisette_full_completion/ htsk_firing_vowel_gisett e/reviewer_table.csv
Spambase	HTSK	0.9287	0.9078	0.0000	0.0000	0.1514	10.15	MAT-9 product-vs- HTSK summary
Spambase	product	0.9270	0.9056	0.0710	0.1218	0.4449	3.84	MAT-9 product-vs- HTSK summary
Vowel	HTSK	0.9444 +/- 0.0156	0.9440 +/- 0.0159	0.0000 +/- 0.0000	0.0000 +/- 0.0000	0.0909 +/- 0.0000	11.0000 +/- 0.0000	/home/harp3133t/ Research/03_Research/ GH-ANFIS_E404/ rebuttal_full_completion/ artifacts/ gisette_full_completion/ htsk_firing_vowel_gisett e/reviewer_table.csv
Vowel	prod	0.9343 +/- 0.0126	0.9339 +/- 0.0128	0.0000 +/- 0.0000	0.0000 +/- 0.0000	0.0909 +/- 0.0000	11.0000 +/- 0.0000	/home/harp3133t/ Research/03_Research/ GH-ANFIS_E404/ rebuttal_full_completion/ artifacts/ gisette_full_completion/ htsk_firing_vowel_gisett e/reviewer_table.csv

Strong Tabular Baselines

dataset	input_dim	RF	HGB	XGBoost	LightGBM	CatBoost	EBM
Breast Cancer	80	0.9467	0.9377	0.9295	0.9381	0.9459	0.9588
Vowel	29	0.9755	0.9143	0.9196	0.9390	0.9031	0.9027
Spambase	57	0.9401	0.9374	0.9375	0.9419	0.9347	0.9345
Gisette	5000	0.9700	0.9733	0.9774	0.9792	0.9732	0.9740

Local Deletion Faithfulness

dataset	grs_intrinsic	shap	lime	interpretation
Breast Cancer	-0.0089	0.1630	0.0687	SHAP strongest under local perturbation
Vowel	0.7333	0.8338	0.6611	all methods respond, SHAP highest
Spambase	0.0693	0.1582	0.0065	SHAP strongest, LIME weak
Gisette	0.00065	0.00065	-0.00001	deletion effect is negligible

Intrinsic Rule Metrics

dataset	mask_stability	primary_coverage	complementary_coverage	top3_rule_agreement
Breast Cancer	1.0000	1.0000	1.0000	0.9928
Vowel	1.0000	1.0000	1.0000	0.9101
Spambase	0.8490	0.6571	1.0000	0.9763
Gisette	0.6011	1.0000	1.0000	0.9998

Paired Statistics

dataset	comparator	acc_delta	acc_p	f1_delta	f1_p
Breast Cancer	ANFIS	+0.0100	0.0249	+0.0144	0.0301
Vowel	GA-ANFIS	+0.1909	0.0019	+0.1944	0.0020
Vowel	PSO-ANFIS	+0.1455	0.0032	+0.1466	0.0036
Vowel	H-ANFIS(Avg)	+0.0273	0.0197	+0.0271	0.0202
Vowel	SVM	+0.1141	<0.001	+0.1152	<0.001
Spambase	GRS-ANFIS(base)	+0.0061	0.0175	+0.0076	0.0216
Spambase	GA-ANFIS	+0.0154	0.0020	+0.0229	<0.001
Spambase	PSO-ANFIS	+0.0352	0.0011	+0.0520	<0.001
Spambase	H-ANFIS(Avg)	+0.0130	0.0026	+0.0188	0.0014
Spambase	SVM	+0.2204	<0.001	+0.3624	<0.001
Gisette	ANFIS	+0.0263	<0.001	+0.0268	<0.001
Gisette	GA-ANFIS	+0.0470	<0.001	+0.0469	<0.001
Gisette	PSO-ANFIS	+0.0527	<0.001	+0.0525	<0.001
Gisette	H-ANFIS(Avg)	+0.0035	0.0146	+0.0036	0.0118